

CS 230 : Discrete Computational Structures

Fall Semester, 2023

HOMEWORK ASSIGNMENT #3

Due Date: Tuesday, September 19

Suggested Reading: Rosen Sections 1.7 - 1.8; Lehman et al. Chapter 1

For the problems below, explain your answers and show your reasoning. You may not get full credit unless you show all your work. Use paragraph-style proofs as shown in class.

1. **[6 Pts]** Prove that for all integers p , p is even if and only if $p^3 + 3$ is odd. *Note: You need to do two proofs, one in each direction. You may not use the theorems shown in class regarding odd and even products.*
2. **[5 Pts]** Prove that if x is irrational then $3x$ is irrational. What proof technique did you use and why?
3. **[6 Pts]** Let m and n be positive integers. Prove that if $mn > 63$, then $m \geq 8$ or $n \geq 10$. What proof technique did you use and why?
4. **[6 Pts]** Let x and y be rational numbers and let z be an irrational number. Prove that $xy + z$ is irrational. Can you use a direct proof? Why or why not?
5. **[6 Pts]** Suppose your student organization has 26 scheduled meetings for the calendar year. Prove that at least three of the meetings fall on the same month. What proof technique do you use?
6. **[6 Pts]** Prove that if $p \geq 5$ or $p \leq -3$ then $(p - 1)^2 \geq 16$. What proof technique did you use and why?
7. **[5 Pts]** Prove that there exist irrational numbers x and y whose sum is rational. Is your proof constructive or non-constructive? Explain.

For more practice, work on the problems from Rosen Sections 1.7 - 1.8; Lehman et al. Chapter 1.